

Technology Stack

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Define Your Technology Stack:

The HDI Predictor System uses modern web development and machine learning technologies to predict the Human Development Index (HDI) based on key development indicators. The selected technology stack provides accurate predictions, a responsive user interface, easy maintenance, and supports future enhancements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, Bootstrap 5, JavaScript	Provides a responsive web interface for entering indicators and viewing HDI prediction results.
2	Backend / Server-Side	Python, Flask	Handles user requests, validates input data, and generates HDI predictions.
3	Machine Learning	Scikit-Learn	Builds and executes the HDI prediction model.
4	Data Processing	Pandas, NumPy	Processes, cleans, and prepares HDI datasets for prediction.
5	Visualization	Matplotlib, Seaborn	Generates charts and exploratory data analysis visualizations.
6	Data Storage	CSV Dataset, Browser LocalStorage	Stores processed HDI data and user prediction history.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
7	Deployment / Hosting	Flask Built-in Server	Runs the application locally for development and demonstration purposes.